



Student Social Media & Mental Health Impact

Multi-class classification using Random Forest & XGBoost

Data Preprocessing

EDA

Model Tuning

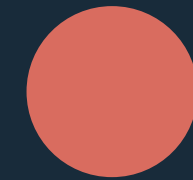
Final Recommendation

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Group : 1
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Predict

survey-based mental health risk levels from students' digital behavior and routine features.



Poor



Fair



Good

5,000

survey responses

13

raw columns

3

risk classes

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04	Model Implementation & Tuning	Random Forest and XGBoost setup with consistent GridSearchCV comparison.
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Problem Definition & Data

Predict mental-health risk category from digital behavior and daily routine features.

Problem statement

Predict whether a student falls into a Poor, Fair, or Good mental health risk level using social media behavior, sleep, study routine, and physical activity.



Dataset snapshot

5,000

raw responses

4,998

after duplicates

80/20

train-test split

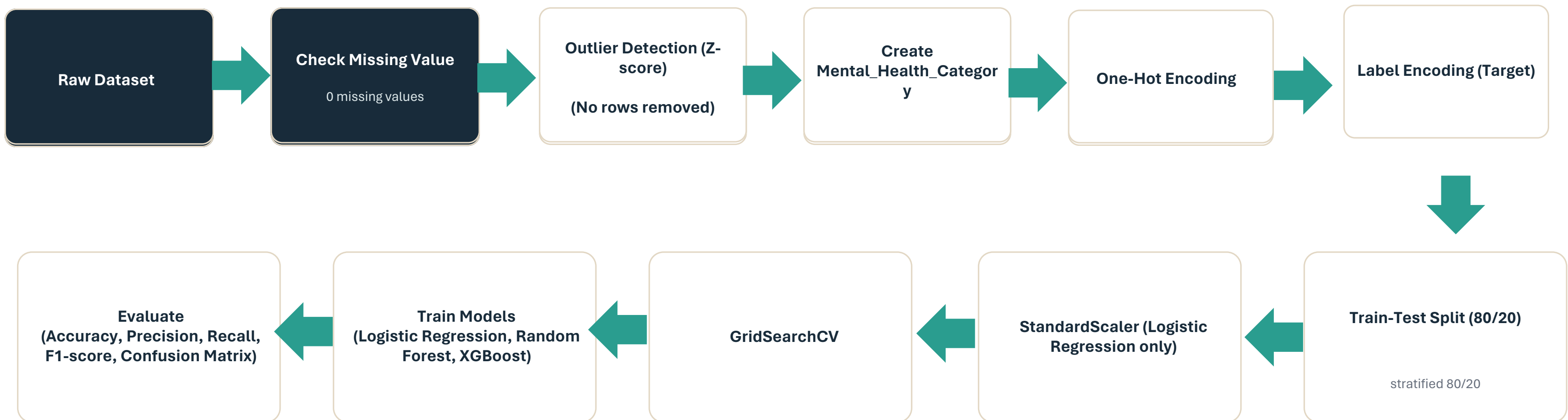
Macro F1

winner metric

Core features

- Daily social media usage hours
- Phone unlock frequency
- Sleep duration per night
- Study hours + physical activity
- Most-used platform + purpose of use

Data Cleaning & Preprocessing



Data Cleaning & Preprocessing

Cleaning steps from the provided notebook

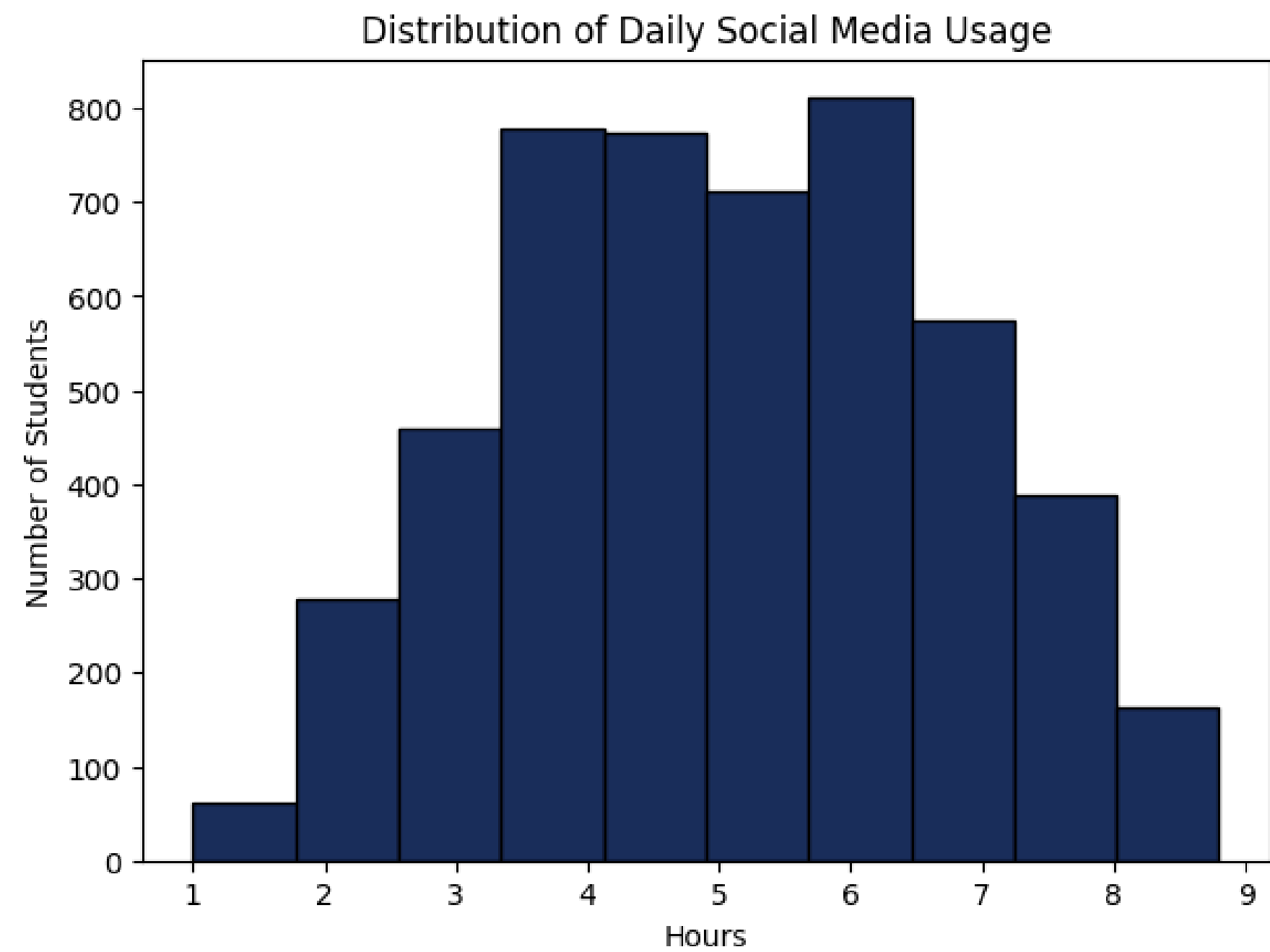
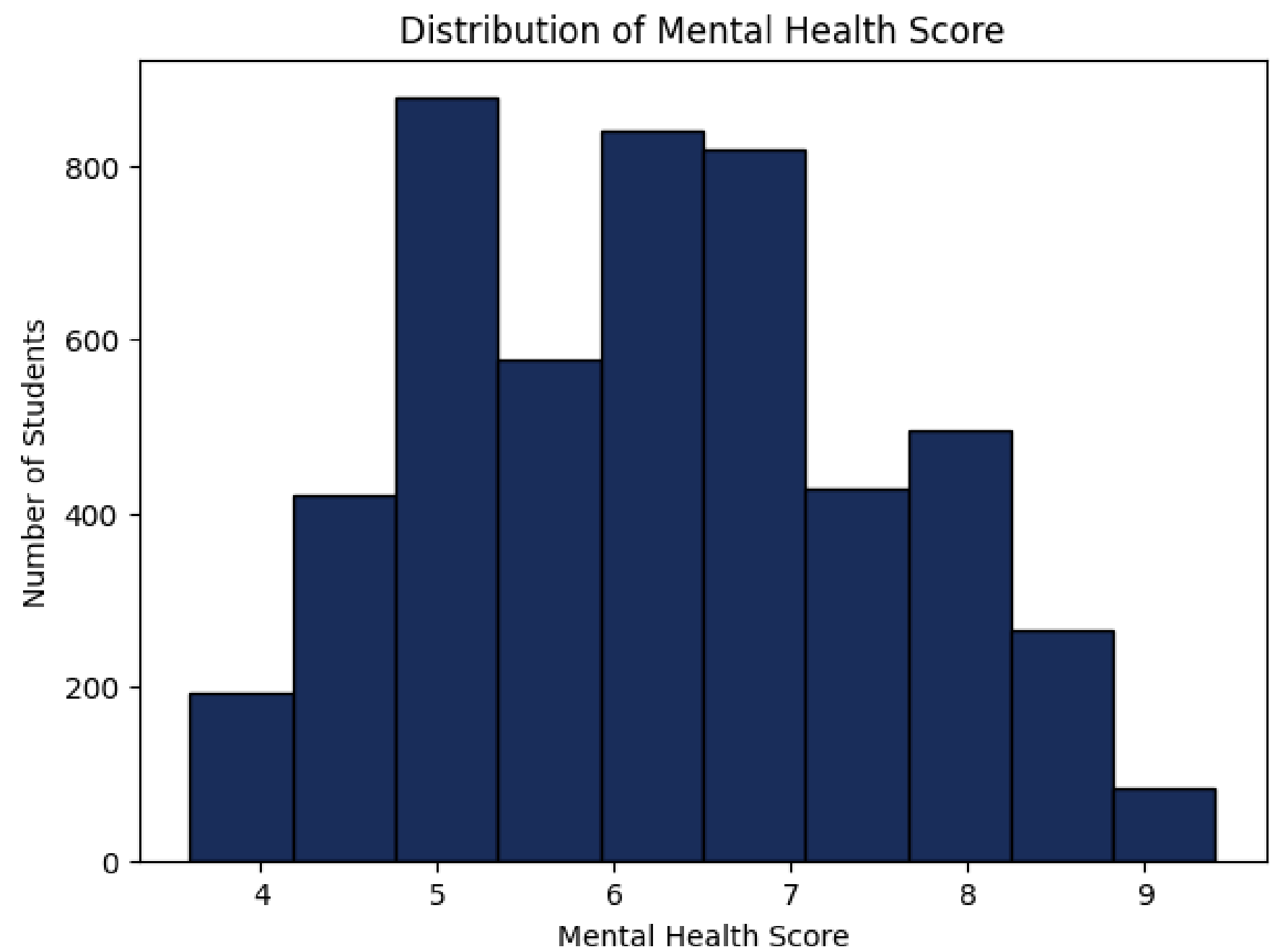
- Checked missing values with `df.isnull().sum()` → no missing values found.
- Counted duplicate rows with `df.duplicated().sum()` → 2 duplicates found.
- Dropped duplicates using `df.drop_duplicates()` → 4,998 rows remained.
- Reviewed outliers using Z-score on Age, Avg Daily Usage, and Sleep Hours.

Model-ready dataset preparation

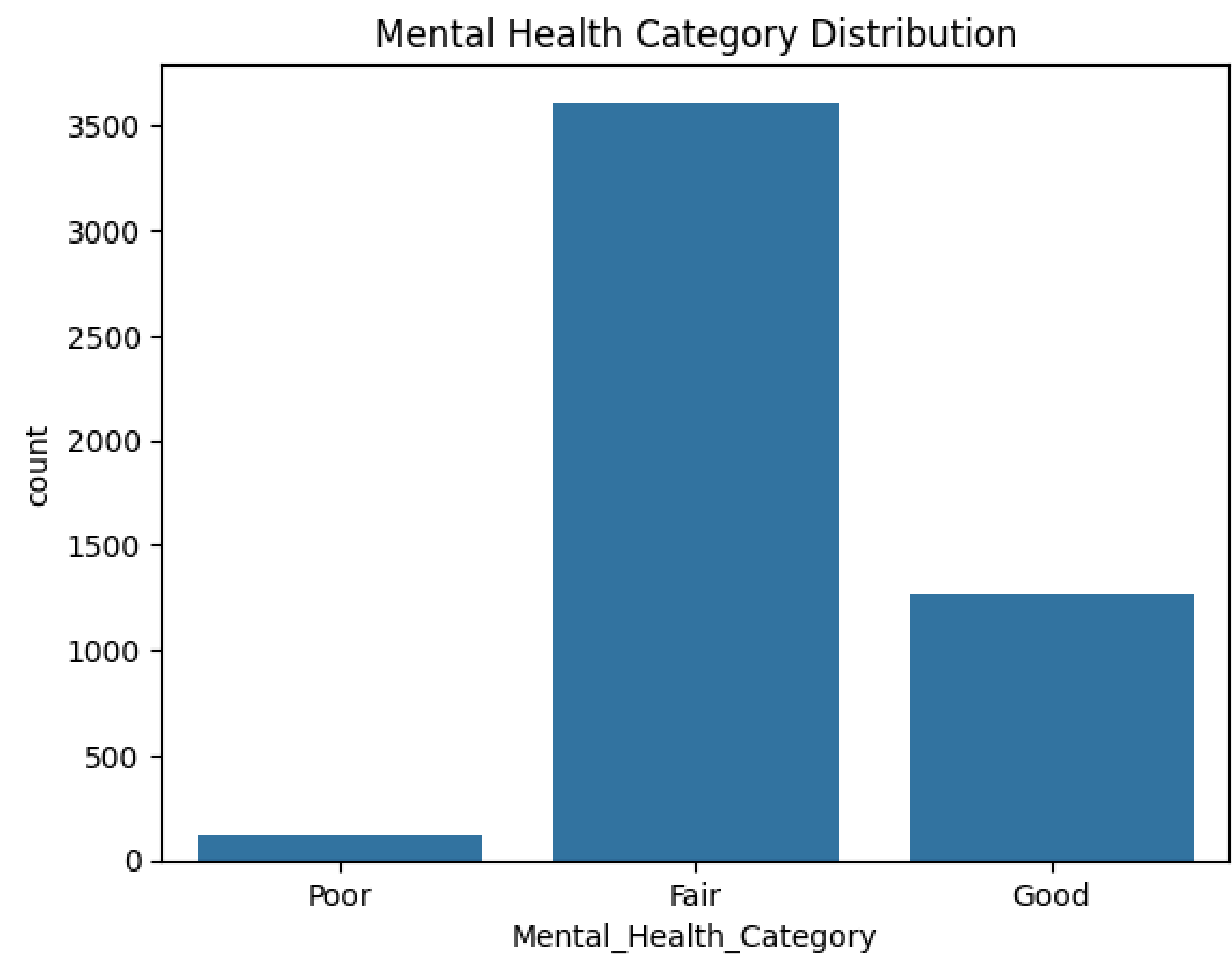
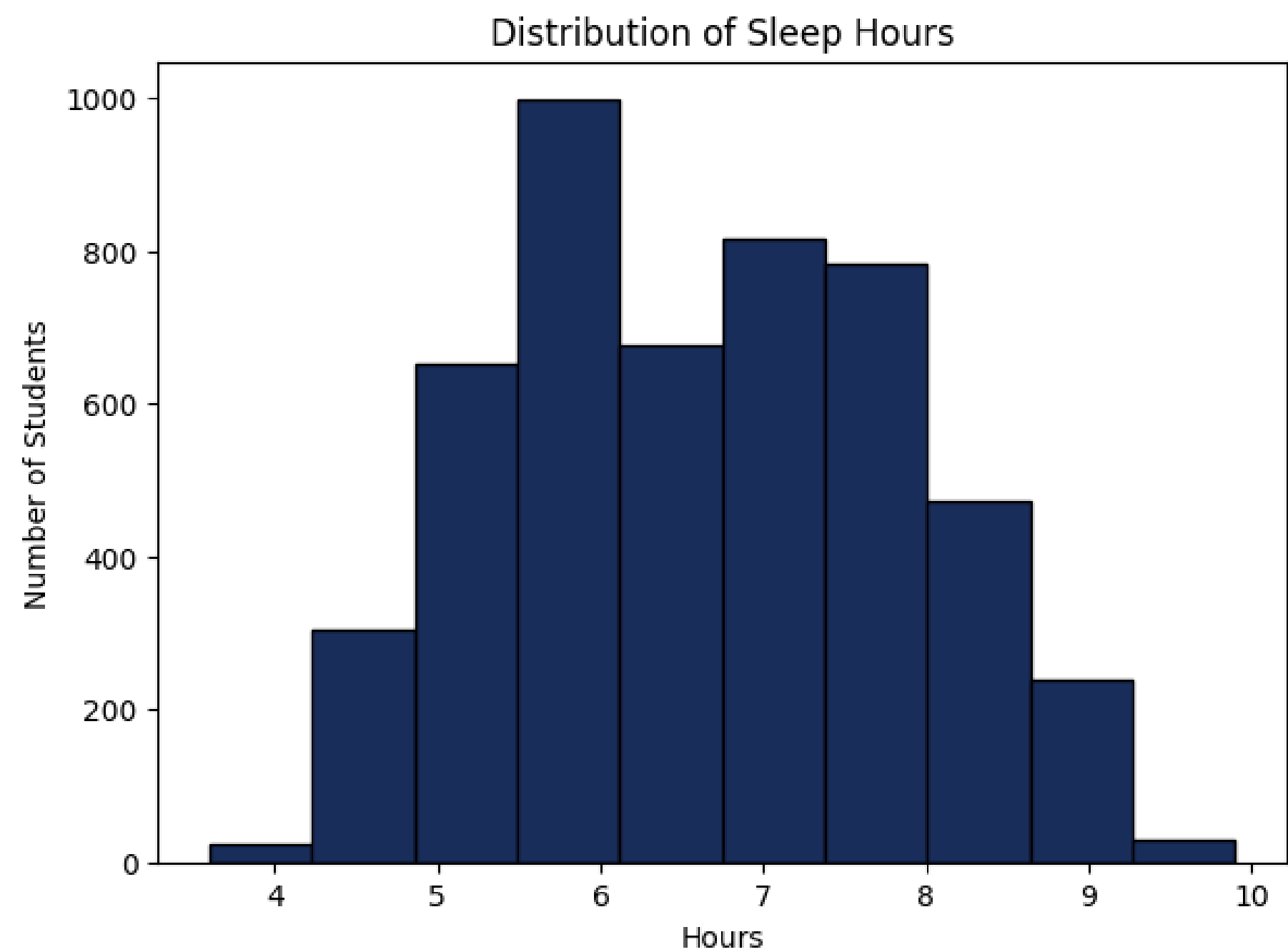
- Binned `Mental_Health_Score` into Poor, Fair, and Good target labels.
- Dropped `Mental_Health_Score`, `Mental_Health_Category`, and `Stress_Level` to reduce leakage.
- Applied one-hot encoding to categorical columns and scaling to numeric columns.
- Used stratified 80/20 train-test split and GridSearchCV scored by macro F1.

Final modeling base: 4,998 student responses after duplicate removal. XGBoost and Random Forest were trained on the same prepared data split for a fair comparison.

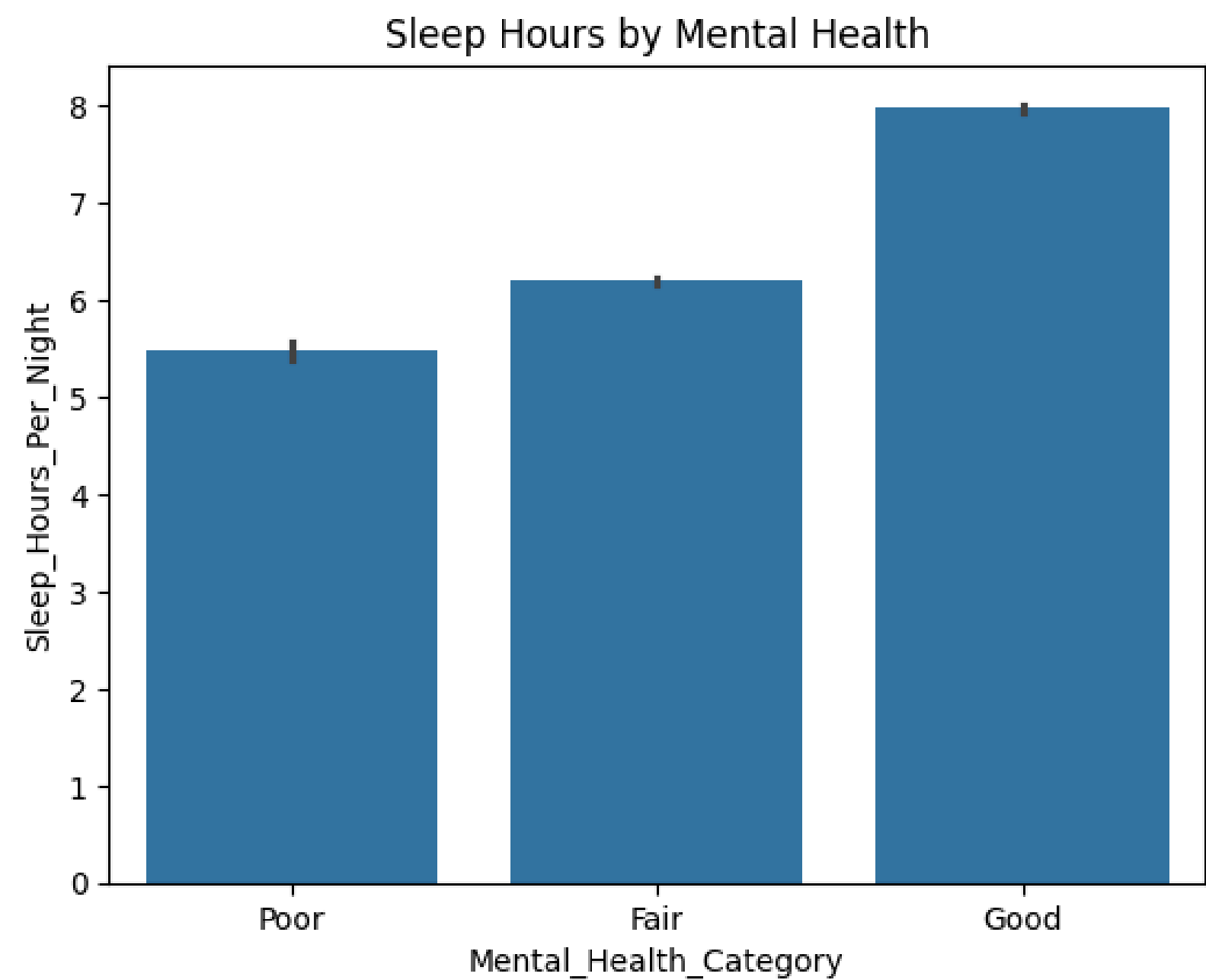
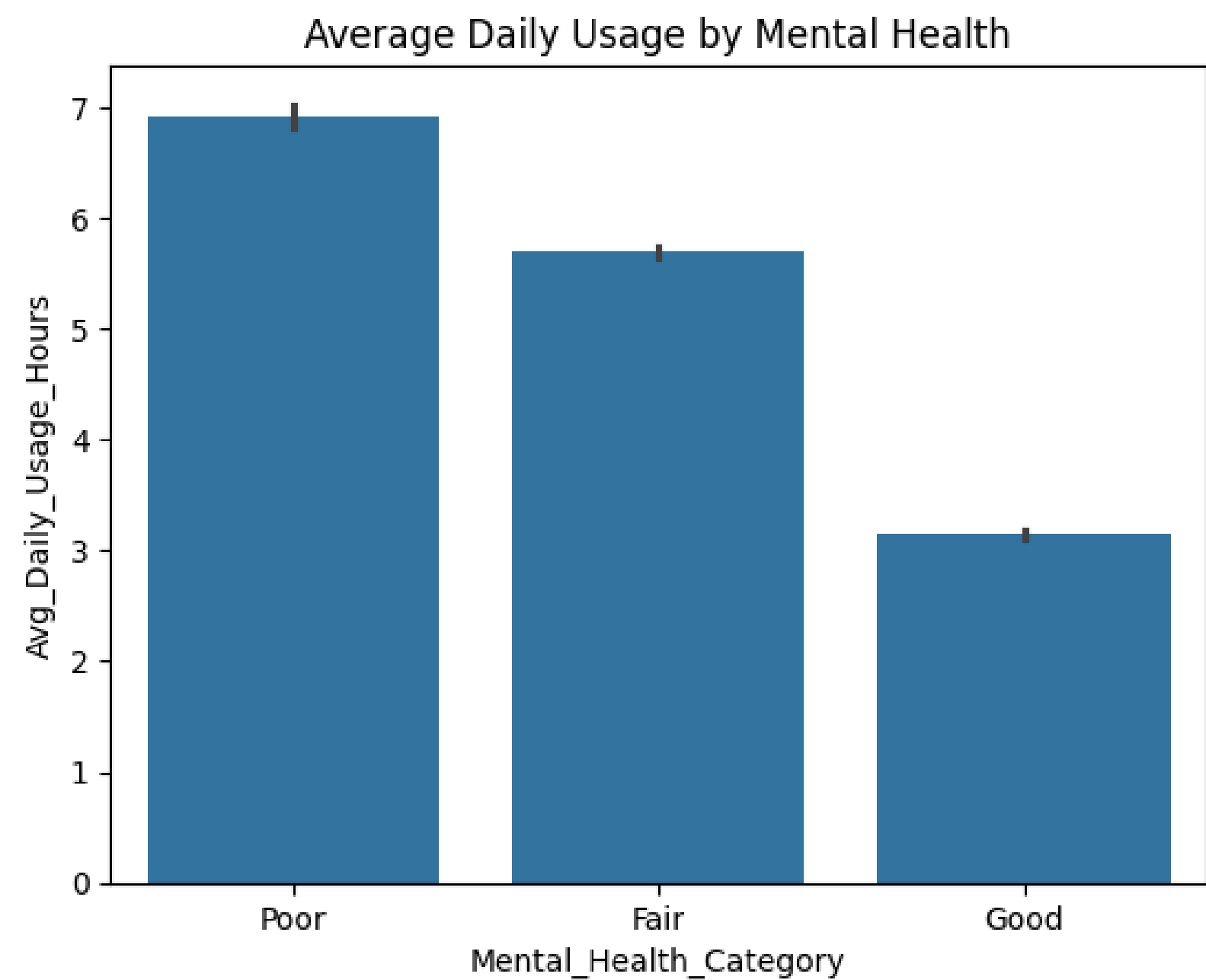
Exploratory Data Analysis: Key Insights



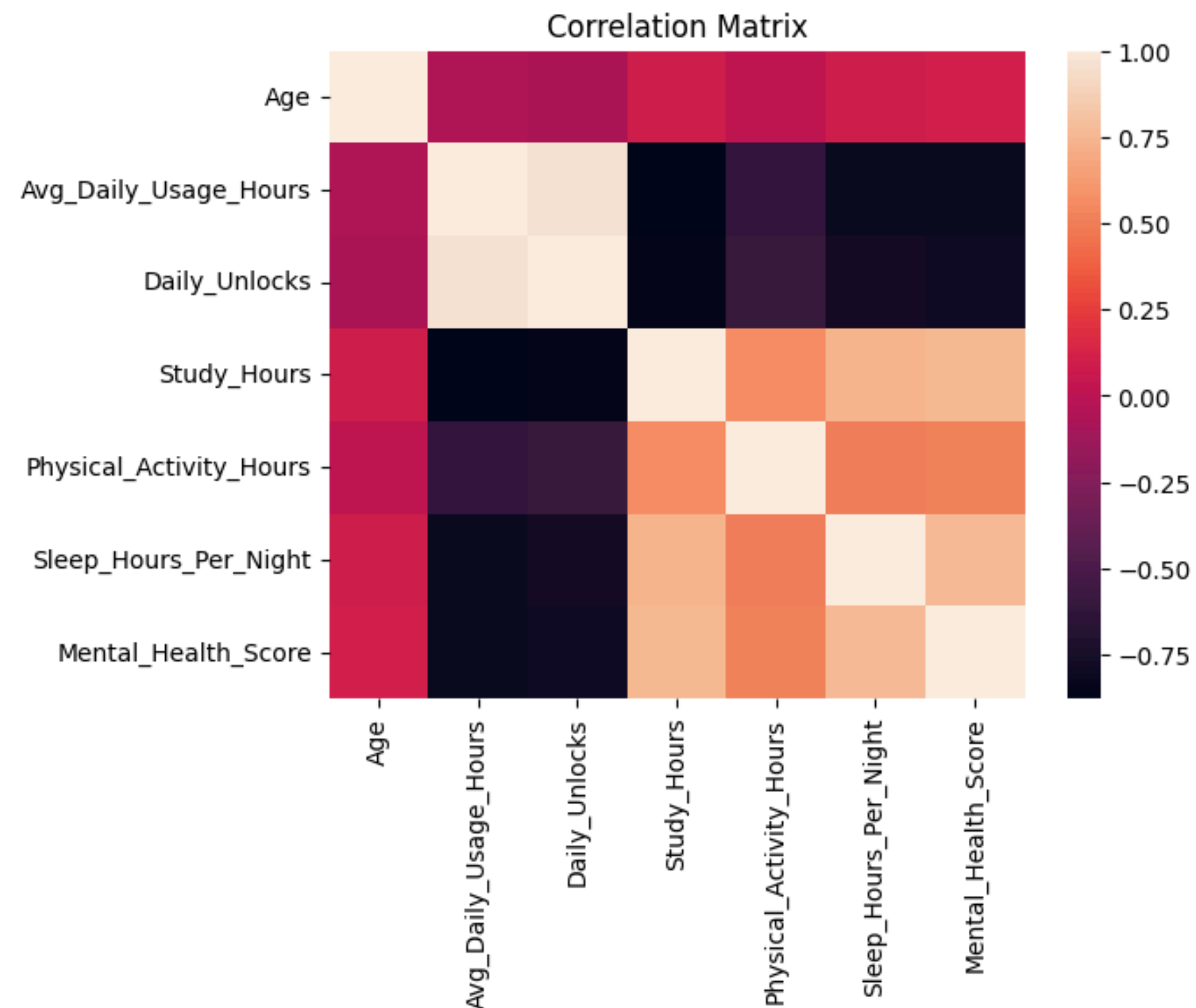
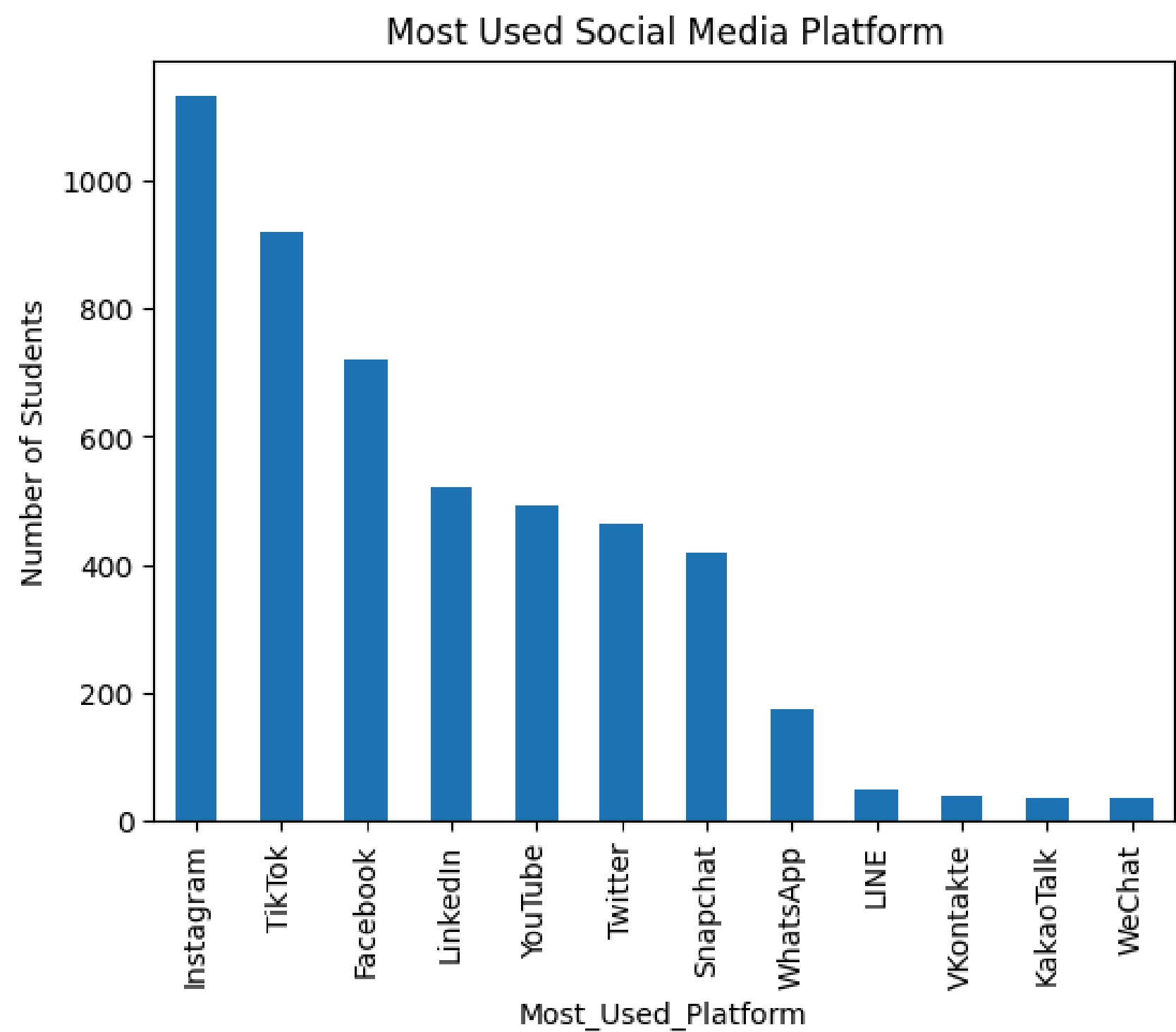
Exploratory Data Analysis: Key Insights



Exploratory Data Analysis: Key Insights



Exploratory Data Analysis: Key Insights



Model Implementation & Tuning



Random Forest

Best parameters

n_estimators = 200
max_depth = None
min_samples_split = 2

Why used: robust baseline, clear feature importance, and strong performance on nonlinear tabular data.

XGBoost

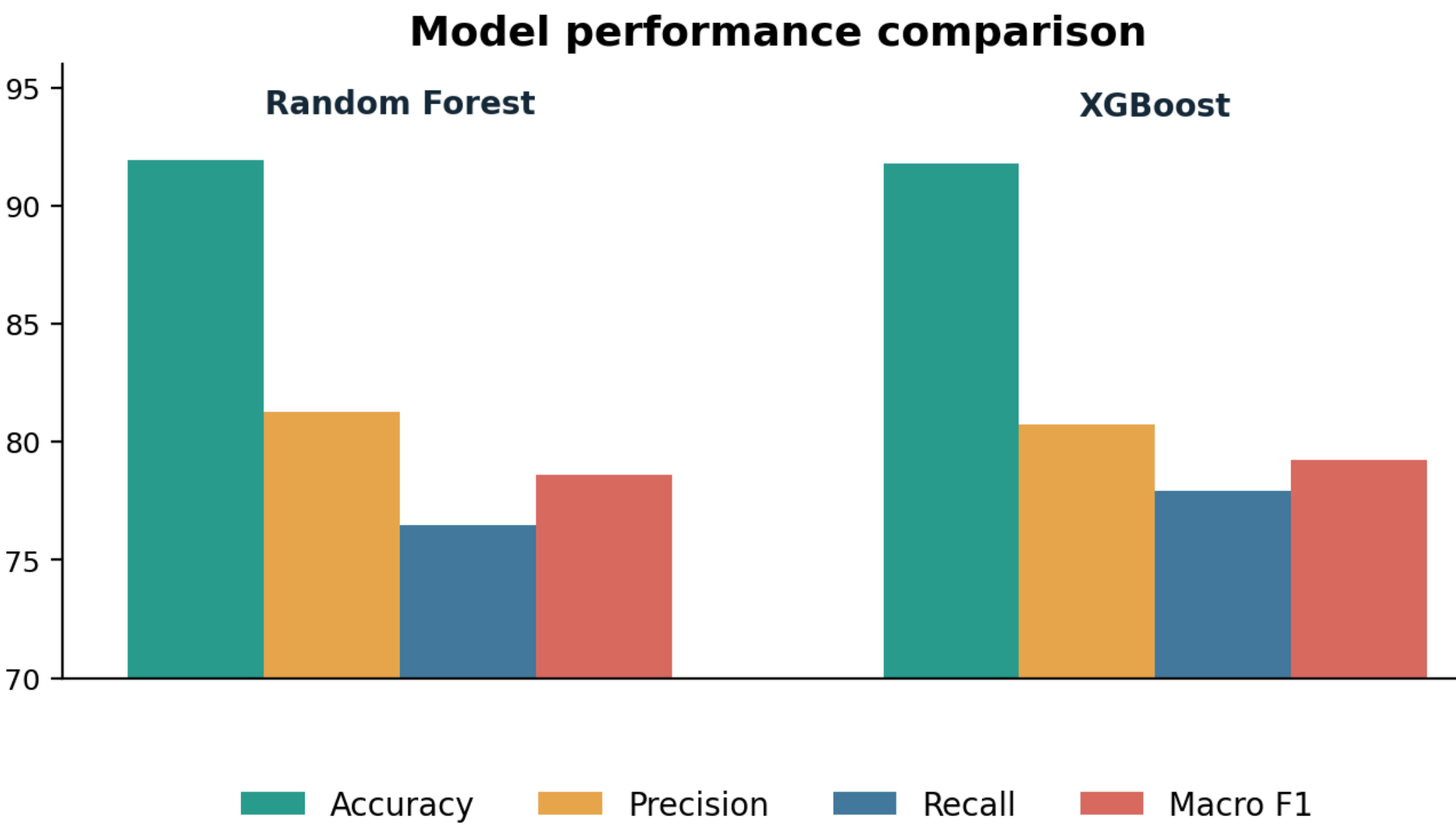
Best parameters

n_estimators = 200
max_depth = 5
learning_rate = 0.1
subsample = 0.8

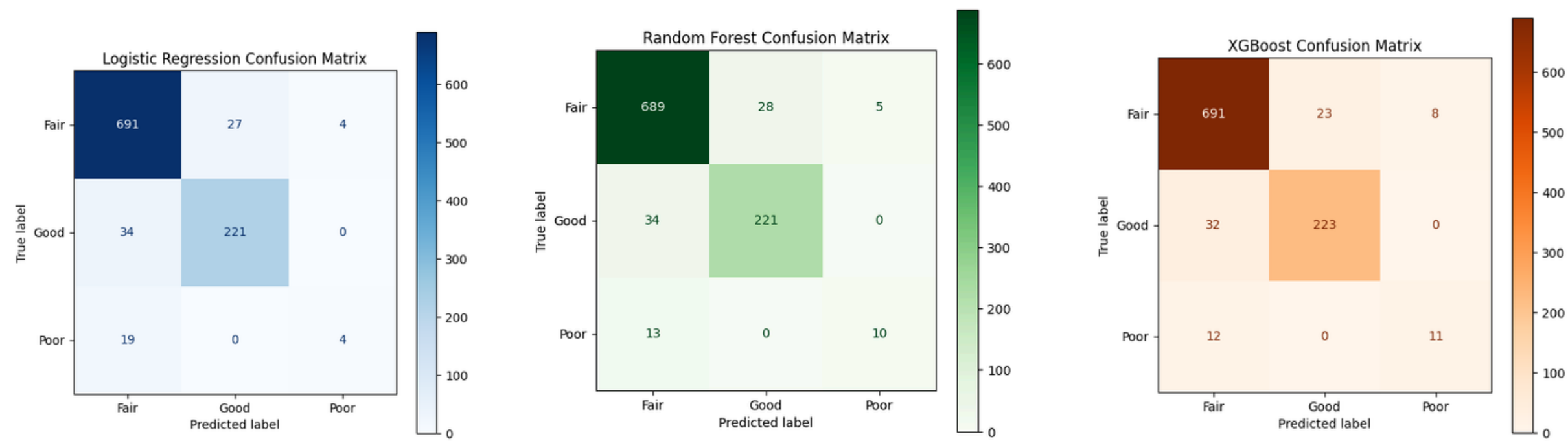
Why used: boosted trees can capture complex interactions and improve minority-class recall.

Metrics: Accuracy • Macro Precision • Macro Recall • Macro F1-score • Confusion Matrix

Results & Model Comparison



Results & Model Comparison



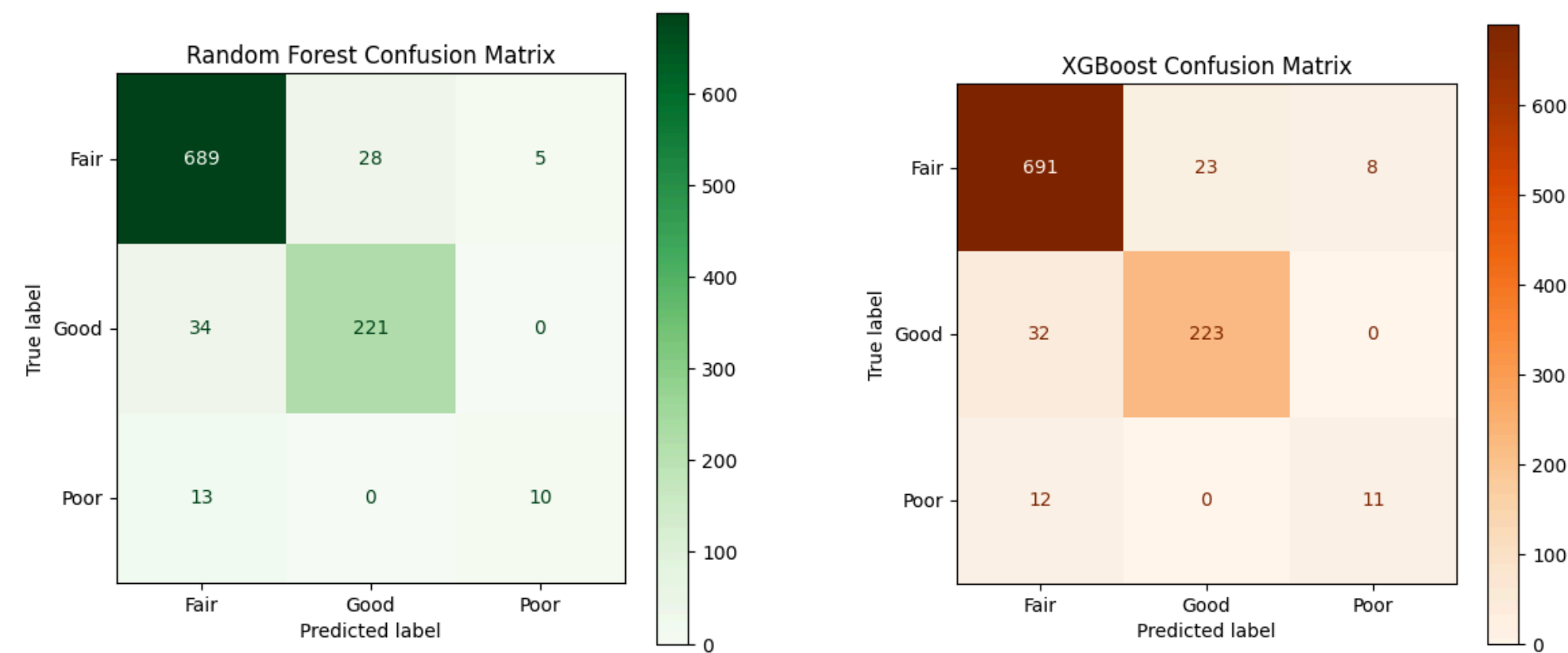
Final metric table

Model	Accuracy	Precision	Recall	F1-score	Poor Recall
Logistic Regression	92.0%	77.0%	67.0%	69.0%	17.0%
Random Forest	92.0%	83.0%	75.0%	78.0%	43.0%
XGBoost	93.0%	81.0%	77.0%	79.0%	48.0%

All main metrics are macro-averaged except Accuracy and Poor Recall.

Winner: XGBoost

Results & Model Comparison



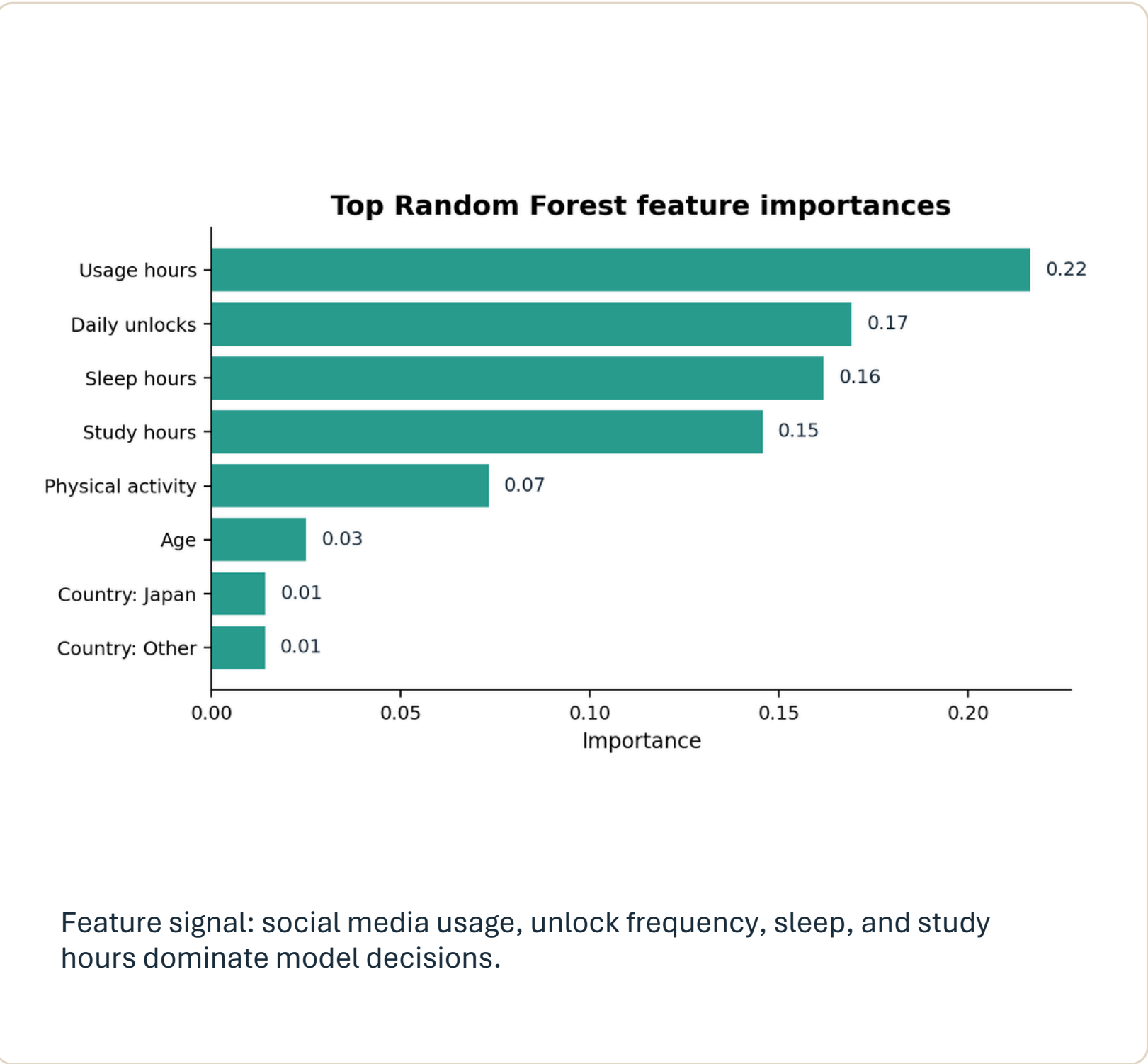
Final metric table

Model	Accuracy	Precision	Recall	F1-score	Poor Recall
Random Forest	91.9%	81.3%	76.5%	78.6%	47.8%
XGBoost	91.8%	80.8%	77.9%	79.2%	52.2%

All main metrics are macro-averaged except Accuracy and Poor Recall.

Winner: XGBoost

Comparative Analysis: Random Forest vs XGBoost



Core comparison		
Dimension	Random Forest	XGBoost
Learning strategy	Bagging: independent trees	Boosting: sequential trees
Strength	Stable and explainable	Better minority-class recall
Tuning	Less sensitive	Needs careful LR/depth tuning
Project fit	Reliable benchmark	Best final candidate

Macro F1 gain: 0.6 pts

Poor recall gain: 4.3 pts

Team Member

Member 1

- Data Preprocessing & Feature Engineering
- Exploratory Data Analysis (EDA)
- Random Forest Classifier implementation & Hyperparameter Tuning

Member 2

- XGBoost Classifier implementation & Hyperparameter Tuning
- Model Comparison & Performance Evaluation
- Summary & Documentation

Conclusion & Next Steps

Final recommendation

After tuning, XGBoost performed best because it achieved the highest macro F1-score and improved recall for the Poor-risk class.

92%

accuracy

79.2%

macro F1

52.2%

Poor recall

Challenges

- Class imbalance, especially Poor risk
- Self-reported survey noise
- Feature correlation: usage, sleep, study
- Avoiding overfitting during tuning

Next steps

- Collect longitudinal behavior data
- Test fairness across student groups
- Tune with class weighting / resampling
- Build a simple risk-awareness dashboard

Thank you